MODEL CARD: HR Turnover Predictor

1. Model Objective

Intended use:
Predict the probability that an employee will leave the company (voluntary or involuntary turnover), in order to help HR teams prioritize preventive retention actions.

Inputs:
A table of individual HR variables (35 features):

Numerical variables: Annual salary, Age, Tenure (years), Engagement score (1–5), Employee satisfaction (1–5), Absences (days/year), Late arrivals over 30 days, Special projects, Absenteeism rate, Engagement risk score

Encoded categorical variables (one-hot): Department, Recruitment source, Performance score, Gender, Marital status

Outputs:

Calibrated probability of leaving between 0 and 1

Predicted class: Active (0) or Resigned (1)

SHAP values per feature to explain individual predictions

2. Training Data

Dataset used:

HR dataset :

- Link: https://www.kaggle.com/datasets/rhuebner/human-resources-data-set
- We tried to use this dataset : https://www.kaggle.com/datasets/carebymanu/employee-dataset-retail

Size and diversity:

- Total samples: 311 employees
- Termd = 0 (Active): 207 (66.6%)
- Termd = 1 (Left): 104 (33.4%)

- Period covered: Hires up to 01/03/2019
- Gender diversity: 53% male, 47% female
- Departments: 6

Excluded variables to prevent data leakage:

- TermReason, EmploymentStatus, EmpStatusID, DateofTermination (post-event information)
- DaysSinceLastReview (proxy of termination date)
- DeptID, PositionID, MaritalStatusID (duplicates of encoded variables)
- Position, State (high cardinality for small dataset)
- CitizenDesc, HispanicLatino, RaceDesc (low predictive value + potential bias)

Anonymized variables (GDPR compliance):

- Employee_Name, EmpID, DOB, Zip removed
- ManagerName pseudonymized (Bayesian target encoding in earlier versions)

3. **Performance**

Metrics used: AUC-ROC, Accuracy, Precision, Recall, F1-score

Evaluation split: 80% train / 20% test (stratified), random seed = 42

Overall results (LightGBM + sigmoid calibration, CV=5):

- AUC-ROC: 0.722
- Accuracy: 0.730

Results by class (test set: 63 employees):

- Active (0): Precision 0.805, Recall 0.786, F1-score 0.795, Support 42
- Resigned (1): Precision 0.591, Recall 0.619, F1-score 0.605, Support 21

Top important features (SHAP / LightGBM importance):

1. Salary
2. EngagementSurvey
3. Age
4. AbsenteeismRate

5. Tenure
6. Absences
7. RiskScore_Engagement
8. EmpSatisfaction
9. MaritalDesc_Single

Data observation:
Counterintuitive pattern: employees who left have higher average tenure (6.7 years) than active ones (5.0 years), suggesting departures mainly involve experienced profiles seeking external growth.

4. Limitations

This model should be considered a prototype, not a production-ready system.

Known limitations

- The dataset is small, with only 311 rows.
- The resigned class remains limited, which makes evaluation less stable.
- Performance may change noticeably with another split or another sample.

Situations not fully covered

- Very recent hires or highly atypical employee profiles.
- Other countries, industries, or labor markets.
- External factors such as economic conditions, competition, or personal events.

Error risks

- Some employees who will leave may not be detected.
- Some employees flagged as risky may not actually resign.

In practice, this means the score should be interpreted as a support signal rather than a definitive judgment.

5. Risks and mitigation

The main risk is over-interpretation. A high turnover score should never be used as the sole basis for an HR decision. Instead, it should trigger human review, contextual analysis, and, when appropriate, preventive discussion with the employee or manager.

Misuse risks

- Using the model as an automated decision-maker.

- Applying it outside the original context without retraining.
- Overreacting to a high-risk score without human validation.

Mitigation measures

- Leakage-prone variables were removed from the final pipeline.
- Predictions are presented as risk indicators, not final decisions.
- SHAP explanations are provided to improve transparency.
- The dashboard is designed to support HR interpretation and retention planning.

There is also a bias risk, especially because some variables may correlate with sensitive demographic patterns. For that reason, any real deployment should include additional fairness checks and clear governance rules.

6. Energy and efficiency

The project was designed to remain lightweight and efficient.

Efficiency characteristics

- Final model: Random Forest.
- CPU-friendly training and inference.
- No GPU required.
- Fast prediction for individual employees in a dashboard setting.

This makes the model well suited for a student project, hackathon demonstration, or internal prototype with limited computational resources.

7. Cybersecurity

The application relies on structured inputs and does not execute dynamic code during prediction. The dataset is loaded locally, and the model does not depend on external APIs for inference.

Security and privacy choices

- Direct personal identifiers were removed from the model features.
- The dashboard relies on EmpID rather than employee names.
- Inputs are constrained through predefined interface components.
- No secrets or API keys are required for core inference.

If the system were deployed in a real HR context, additional protections would still be necessary, especially:

- role-based access control,

- secure storage of datasets and model files,
- audit logging for prediction access and usage.

This model is therefore best understood as a transparent and lightweight HR analytics prototype, designed to assist human judgment rather than replace it.

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